

Decoherence, Informational Entropy, and the Origin of a Dual-Sector Gravitational Coupling in the μ – Σ Framework

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Abstract

We present a physical argument connecting gravitational decoherence, Landauer’s principle, and horizon thermodynamics to the phenomenological μ – Σ modified gravity framework. The argument proceeds in four steps, each grounded in established physics: (i) gravitational structure formation is the dominant source of quantum decoherence in cosmology, producing irreversible classical information at a rate set by the virial theorem; (ii) each bit of classical information carries a minimum thermodynamic cost $kT \ln 2$ (Landauer’s bound), which is realized at the cosmological horizon temperature; (iii) the cumulative informational entropy is encoded on the horizon (holographic principle), modifying the entropy functional that enters the Jacobson–Cai–Kim thermodynamic derivation of the gravitational field equations; (iv) the modification applies exclusively to degrees of freedom on timelike worldlines, because decoherence requires nonzero proper time, while null degrees of freedom (photons) are unaffected. These four steps yield a specific prediction within the standard μ – Σ parameterization: $\mu(a) < 1$ for matter perturbations and $\Sigma(a) = 1$ for photon propagation. The predicted amplitude $\mu_0 = -0.135$ is derived from the virial coupling $\beta = \Omega_m/2$ and the activation function $\mathcal{E}(a) = \exp(1 - 1/a)$. A companion paper demonstrates that this prediction is consistent with Planck 2018 and large-scale structure data ($\Delta\chi^2 = +1.34$). The framework makes no appeal to new fields, extra dimensions, or modifications to the gravitational action; the sole new input is the identification of decoherence-produced information as a contribution to horizon entropy.

1 Introduction

The origin of the arrow of time, the thermodynamic interpretation of gravity, and the information-theoretic content of black hole entropy are typically treated as distinct problems in theoretical physics. In this paper we argue that they are connected through a single physical process—quantum decoherence—and that the connection has observable cosmological consequences.

The standard account of the arrow of time appeals to the second law of thermodynamics: entropy increases, and this asymmetry defines the direction of time [Penrose, 1989]. However, the microscopic laws of physics are time-reversal invariant. A more fundamental account identifies decoherence as the origin of irreversibility [Zurek, 2003, Schlosshauer, 2007]. When a quantum system interacts with an environment, information about the system’s state becomes entangled with environmental degrees of freedom. This process is irreversible: recovering the coherent superposition requires accessing all environmental correlations, which is thermodynamically prohibited

by Landauer’s principle [Landauer, 1961]—erasing one bit of information dissipates at least $kT \ln 2$ of energy. The arrow of time is therefore the arrow of information production.

Independently, Jacobson [Jacobson, 1995] demonstrated that the Einstein field equations can be derived as a thermodynamic equation of state by applying the Clausius relation $\delta Q = T dS$ to local causal horizons, provided the entropy is proportional to the horizon area (the Bekenstein–Hawking relation [Bekenstein, 1973, Hawking, 1975]). Cai and Kim [Cai & Kim, 2005] extended this to the Friedmann–Robertson–Walker (FRW) cosmological apparent horizon, recovering the Friedmann equations from the first law of thermodynamics.

The key feature of Jacobson’s result is its generality: *changing the entropy functional changes the field equations*. If additional entropy contributions are present on cosmological horizons, the Friedmann equations are modified accordingly.

In this work we identify a specific additional entropy contribution: the cumulative classical information produced by gravitational decoherence during structure formation. We show that this identification, combined with the causal structure of spacetime, predicts a dual-sector gravitational coupling in which matter perturbations experience a suppressed effective Newton’s constant ($\mu < 1$) while photon propagation remains standard ($\Sigma = 1$). The prediction is expressed within the μ – Σ framework [Pogosian & Silvestri, 2016] and is quantitatively testable with current and upcoming data.

2 Decoherence and the Arrow of Time

We briefly review the connection between decoherence, information production, and thermodynamic irreversibility, following Zurek [2003] and Schlosshauer [2007].

2.1 Quantum Decoherence

Consider a quantum system $\mathcal{S}$ with Hilbert space $\mathcal{H}_S$ coupled to an environment $\mathcal{E}$ with Hilbert space $\mathcal{H}_E$. An initial product state $|\psi_S\rangle \otimes |E_0\rangle$ evolves under the total Hamiltonian into an entangled state:

$$|\psi_S\rangle \otimes |E_0\rangle \longrightarrow \sum_i c_i |s_i\rangle \otimes |E_i\rangle, \quad (1)$$

where $\{|s_i\rangle\}$ are the pointer states selected by the system–environment interaction. When the environmental states become approximately orthogonal ($\langle E_i | E_j \rangle \approx \delta_{ij}$), the reduced density matrix of $\mathcal{S}$ becomes diagonal:

$$\rho_S = \text{Tr}_E |\Psi\rangle \langle \Psi| \longrightarrow \sum_i |c_i|^2 |s_i\rangle \langle s_i|. \quad (2)$$

Coherence is lost. The system has transitioned from a quantum superposition to a classical mixture. This transition is irreversible in practice because recovering the off-diagonal elements requires accessing all environmental correlations.

2.2 Information Production and Landauer’s Bound

Each decoherence event produces classical information: the specification of which pointer state $|s_i\rangle$ was selected. For a system decohering into N distinguishable states, the information content is

$$I = \log_2 N \quad \text{bits}. \quad (3)$$

Landauer's principle [Landauer, 1961] establishes that erasing one bit of information requires a minimum energy dissipation of

$$\Delta E_{\text{Landauer}} = kT \ln 2 \quad (4)$$

at temperature T . This bound has been experimentally verified [Bérut et al., 2012, Jun et al., 2014] and is now understood as a fundamental connection between information theory and thermodynamics.

The irreversibility of decoherence is therefore not merely practical but thermodynamic: undoing the decoherence would require erasing the information produced, which would dissipate energy into the environment, increasing its entropy. The net entropy of the universe increases with each decoherence event. This is the physical content of the arrow of time.

3 Gravity as Horizon Thermodynamics

We review the Jacobson [Jacobson, 1995] and Cai–Kim [Cai & Kim, 2005] derivations to establish the framework within which the informational entropy enters.

3.1 Jacobson's Derivation

At any spacetime point p , the equivalence principle guarantees a locally flat neighborhood. Consider a local Rindler horizon $\mathcal{H}$ generated by an approximate boost Killing vector χ^a . The Unruh temperature [Unruh, 1976] experienced by an accelerating observer is

$$T = \frac{\hbar a_{\perp}}{2\pi c k_B}, \quad (5)$$

where $a_{\perp}$ is the acceleration perpendicular to $\mathcal{H}$.

The heat flux through $\mathcal{H}$ is

$$\delta Q = \int_{\mathcal{H}} T_{ab} \chi^a d\Sigma^b, \quad (6)$$

where T_{ab} is the stress-energy tensor. The entropy is taken proportional to the horizon area:

$$S = \frac{\eta A}{4 \ell_P^2}, \quad (7)$$

where $\eta = 1$ for standard Bekenstein–Hawking entropy and ℓ_P is the Planck length.

The area change is related to the expansion θ of the null generators via the Raychaudhuri equation:

$$\frac{d\theta}{d\lambda} = -\frac{1}{2}\theta^2 - \sigma_{ab}\sigma^{ab} - R_{ab}k^ak^b, \quad (8)$$

where k^a is the null tangent vector and σ_{ab} is the shear.

Imposing the Clausius relation $\delta Q = T dS$ for all local causal horizons, and using the Raychaudhuri equation to relate the area change to the Ricci tensor, Jacobson obtained:

$$R_{ab} - \frac{1}{2} R g_{ab} + \Lambda g_{ab} = \frac{8\pi G}{c^4} T_{ab}. \quad (9)$$

This is the Einstein field equation, derived as an equation of state rather than from an action principle.

3.2 The Generality of the Derivation

Jacobson [Jacobson, 1995] noted that modifying the entropy functional modifies the resulting field equations. If S depends on curvature invariants, higher-derivative gravity theories emerge. The derivation is a machine: *specify an entropy, derive a gravity theory*.

3.3 Cai–Kim: The Friedmann Equations

Cai and Kim [Cai & Kim, 2005] applied this logic to the FRW apparent horizon. For a spatially flat FRW universe with line element $ds^2 = -dt^2 + a^2(t) d\mathbf{x}^2$, the apparent horizon is at

$$\tilde{r}_A = \frac{1}{H}, \quad (10)$$

with area $A_H = 4\pi/H^2$, geometric entropy

$$S_{\text{geo}} = \frac{\pi}{G H^2}, \quad (11)$$

and temperature

$$T_H = \frac{H}{2\pi}. \quad (12)$$

Applying the first law $-dE = T_H dS_{\text{geo}}$ (where E is the Misner–Sharp energy within the horizon), Cai and Kim recovered the Friedmann equations:

$$H^2 = \frac{8\pi G}{3} \rho, \quad (13)$$

$$\dot{H} = -4\pi G (\rho + P), \quad (14)$$

with Λ entering as an integration constant.

The derivation confirms Jacobson’s result in the cosmological setting: the Friedmann equations are the thermodynamic equation of state of the FRW apparent horizon.

4 Informational Entropy from Gravitational Decoherence

We now identify the additional entropy contribution that modifies the Friedmann equations.

4.1 Gravitational Structure Formation as Decoherence

In the standard cosmological model, primordial density perturbations grow under gravitational instability. As overdensities enter the nonlinear regime, they collapse, virialize, and form bound structures (halos, galaxies, clusters). This process involves the irreversible production of classical information: each virialized structure represents a definite classical configuration selected from the space of possible quantum states.

The decoherence rate is governed by the gravitational interaction strength. In a cosmological context, the relevant timescale is the Hubble time H^{-1} , and the relevant mass scale is the total matter content within the horizon. The rate of information production per unit horizon area, at the horizon temperature $T_H = H/(2\pi)$, defines the informational entropy production rate.

4.2 The Virial Coupling

The fraction of gravitational energy that is converted to irreversible (thermalized, decohered) degrees of freedom during structure formation is determined by the virial theorem. For a self-gravitating system in virial equilibrium,

$$2K + U = 0, \quad (15)$$

where K is the kinetic energy and U is the gravitational potential energy. The total energy is $E_{\text{tot}} = K + U = U/2$. Half of the gravitational potential energy is converted into random (thermal) kinetic energy of the constituent particles; the other half remains as bulk gravitational binding energy.

The thermalized fraction is the fraction that produces decoherence. The coupling of the informational entropy to the horizon thermodynamics is therefore

$$\beta = \frac{\Omega_m}{2}, \quad (16)$$

where Ω_m is the matter density parameter. For $\Omega_m = 0.315$, this gives $\beta = 0.1575$.

4.3 The Activation Function

The cumulative nature of information production is encoded in the activation function. Each bit of classical information produced by decoherence is encoded on the cosmological horizon and is not subsequently erased (information is conserved on the horizon in the absence of a mechanism for global erasure).

The information production rate per unit horizon area scales as H/a : the expansion rate H sets the causal timescale over which decoherence occurs, while the factor $1/a$ accounts for the dilution of the comoving density of decohering structures. The cumulative informational entropy is then

$$\mathcal{S}_{\text{info}}(a) = \int_0^a \frac{H}{a'} da' \propto 1 - \frac{1}{a} + \text{const.} \quad (17)$$

The constant is fixed by the boundary condition $\mathcal{S}_{\text{info}}(a \rightarrow 0) = 0$ (no structures, no information).

The contribution to the horizon's thermodynamic potential is obtained by exponentiating the entropy, as appropriate for a partition function: each bit doubles the number of distinguishable microstates on the horizon (N bits $\rightarrow 2^N = e^{N \ln 2}$ microstates). The activation function is therefore

$$\mathcal{E}(a) = \exp(\mathcal{S}_{\text{info}}) = \exp\left(1 - \frac{1}{a}\right). \quad (18)$$

This function has the required properties:

- $\mathcal{E}(a) \rightarrow 0$ as $a \rightarrow 0$: no structures at early times, no informational entropy, no modification.
- $\mathcal{E}(a) = 1$ at $a = 1$: full activation at the present epoch.
- $\mathcal{E}(a) \approx 0$ at the epoch of recombination ($a \approx 10^{-3}$): early-universe physics is preserved.

5 Modified Friedmann Equation and the Dual-Sector Split

5.1 Modified First Law

With the total horizon entropy

$$S_{\text{total}} = S_{\text{geo}} + S_{\text{info}}, \quad (19)$$

the Cai–Kim first law becomes

$$-dE = T_H d(S_{\text{geo}} + S_{\text{info}}). \quad (20)$$

The geometric piece yields the standard Friedmann equation as before. The informational piece yields an additional contribution:

$$H^2 = \frac{8\pi G}{3} \rho + \frac{\Lambda}{3} + \beta \mathcal{E}(a) H_0^2, \quad (21)$$

where $\beta = \Omega_m/2$ and $\mathcal{E}(a) = \exp(1 - 1/a)$.

5.2 The Causal Origin of the Dual-Sector Split

The argument above applies to degrees of freedom that undergo gravitational decoherence. We now show that the causal structure of spacetime determines which degrees of freedom are affected.

Definition 1 (Decoherence eligibility). *A degree of freedom undergoes gravitational decoherence if and only if it propagates on a timelike worldline, i.e., it experiences nonzero proper time $d\tau > 0$.*

The physical basis for this definition is as follows. Decoherence is a dynamical process: the system–environment interaction in Eq. (1) requires time to produce orthogonalization of the environmental states ($\langle E_i | E_j \rangle \rightarrow \delta_{ij}$). The decoherence timescale τ_D is finite and positive. For a degree of freedom with proper time interval $\Delta\tau$, decoherence occurs when $\Delta\tau \gg \tau_D$.

For massive particles (baryons, dark matter), $\Delta\tau > 0$ along their worldlines. They interact gravitationally over cosmological timescales, cluster, virialize, and decohere. They produce informational entropy.

For photons, the worldline is null: $ds^2 = 0$, and the proper time is $\Delta\tau = 0$. A photon does not experience duration. It does not undergo a sequence of interactions over time. It does not decohere through gravitational clustering. It produces no informational entropy through this channel.

Proposition 1 (Dual-sector coupling). *Let S_{info} denote the informational entropy from gravitational decoherence. Then:*

1. *For matter (timelike worldlines): $S_{\text{info}} > 0$, and the modified first law yields a suppressed gravitational coupling $\mu(a) < 1$.*
2. *For photons (null worldlines): $S_{\text{info}} = 0$, and the standard first law yields $\Sigma(a) = 1$.*

The proof follows directly from Definition 1 and the Cai–Kim derivation. When $S_{\text{info}} = 0$, the first law reduces to $-dE = T_H dS_{\text{geo}}$, which yields standard GR. When $S_{\text{info}} > 0$, the additional entropy modifies the field equations for the matter sector only.

5.3 Mapping to the μ - Σ Framework

The μ - Σ parameterization [Pogosian & Silvestri, 2016] encodes deviations from GR through the modified Poisson equations:

$$k^2 \Phi = -4\pi G \mu(a) a^2 \bar{\rho}_m \delta_m, \quad (22)$$

$$k^2(\Phi + \Psi) = -8\pi G \Sigma(a) a^2 \bar{\rho}_m \delta_m. \quad (23)$$

Proposition 1 maps directly onto this framework:

- **Matter sector** (μ): The informational entropy modifies the gravitational coupling governing density perturbations. The effective coupling is

$$\mu(a) = \frac{H_{\Lambda\text{CDM}}^2(a)}{H_{\Lambda\text{CDM}}^2(a) + \beta \mathcal{E}(a)}, \quad (24)$$

which gives $\mu(a) < 1$ at late times and $\mu(a) \rightarrow 1$ at early times. At $a = 1$:

$$\mu_0 \equiv \mu(z=0) - 1 = \frac{-\beta}{1 + \beta} = -0.135. \quad (25)$$

- **Photon sector** (Σ): Photon geodesics are determined by the lensing potential $(\Phi + \Psi)/2$, which depends on null-worldline thermodynamics. Since $S_{\text{info}} = 0$ for null degrees of freedom, the lensing potential is unmodified:

$$\Sigma(a) = 1 \quad (\text{exact}). \quad (26)$$

This is the central prediction: $\mu < 1$, $\Sigma = 1$. The suppression $\mu < 1$ means that matter perturbations grow more slowly than in GR. The preservation $\Sigma = 1$ means that CMB anisotropies, gravitational lensing, and BAO angular positions are unaffected.

6 Observational Consequences

The prediction $\mu_0 = -0.135$, $\Sigma_0 = 0$ has been tested in a companion paper [Mahaffey, 2026] using twelve MCMC analyses with Planck 2018 and large-scale structure data. We summarize the results here.

6.1 Consistency with Current Data

The fixed prediction $\mu_0 = -0.135$ yields:

- Planck only: $\Delta\chi^2 = +1.43$ relative to ΛCDM .
- Planck + RSD: $\Delta\chi^2 = +1.34$ relative to ΛCDM .

Both values are below the 1σ threshold for a model with the same number of free parameters. The prediction is not excluded by current data.

6.2 The σ_8 Shift

The suppressed coupling reduces the growth of density perturbations, shifting σ_8 from 0.813 (Λ CDM) to 0.800 (fixed $\mu_0 = -0.135$). This 1.6% reduction arises from the weakened Poisson source term and is determined entirely by the predicted amplitude; it is not fitted to any late-time data.

The magnitude and direction of this shift are consistent with the reported discrepancy between CMB-derived and weak-lensing-derived values of σ_8 [Heymans et al., 2021, Abbott et al., 2022]. A quantitative assessment requires a joint analysis with weak lensing likelihoods, which is deferred to future work.

6.3 Photon-Sector Observables

As predicted by $\Sigma = 1$:

- CMB TT, TE, EE power spectra are unchanged.
- CMB lensing is unchanged to within 0.3%.
- BAO angular positions are unchanged.
- Supernova luminosity distances are unchanged at the perturbation level.

6.4 Forecast Sensitivity

Published Euclid and DESI forecasts [Frusciante et al., 2025] project $\sigma(\mu_0) \sim 0.03\text{--}0.05$, sufficient to test $\mu_0 = -0.135$ at $> 3\sigma$ significance.

7 Discussion

7.1 Summary of the Argument

The logical chain is:

1. Gravitational structure formation produces quantum decoherence.
2. Decoherence produces irreversible classical information (Zurek).
3. Information erasure has a minimum thermodynamic cost (Landauer).
4. The cumulative information is encoded on the cosmological horizon (holographic principle).
5. The encoding rate is set by the virial theorem: $\beta = \Omega_m/2$.
6. The cumulative encoding modifies the horizon entropy: $S_{\text{total}} = S_{\text{geo}} + S_{\text{info}}$.
7. Modified entropy $\rightarrow$ modified gravitational coupling for matter: $\mu < 1$ (Jacobson–Cai–Kim).
8. Photons do not decohere (null worldlines, $\Delta\tau = 0$) $\rightarrow \Sigma = 1$.
9. The net effect is a 13.5% suppression of the effective gravitational coupling at $z = 0$.
10. The suppression vanishes at early times because $\mathcal{E}(a) \rightarrow 0$.

Each step uses established physics except step 6, which is the sole new identification: decoherence-produced information contributes to horizon entropy.

7.2 Relation to Previous Work

Jacobson’s derivation [Jacobson, 1995] established that the Einstein equations are an equation of state. Verlinde [Verlinde, 2011] and Padmanabhan [Padmanabhan, 2010] extended this perspective to entropic gravity. The present work differs in that it does not modify the gravitational dynamics globally; the modification arises only for the matter sector and is sourced by a specific physical process (gravitational decoherence) rather than by a general entropic force.

The μ – Σ framework has been used extensively in observational analyses [Pogosian & Silvestri, 2016, Abbott et al., 2023, Andrade et al., 2024, DESI Collaboration, 2025]. However, existing theoretical motivations for specific μ – Σ values come from scalar-tensor theories, $f(R)$ gravity, and braneworld models, all of which generically predict $\mu \geq 1$. The present framework provides, to our knowledge, the first physical motivation for $\mu < 1$ with $\Sigma = 1$.

7.3 Assumptions and Limitations

The argument rests on the following assumptions, which we state explicitly:

1. **Gravity is a thermodynamic equation of state.** This is the Jacobson hypothesis. It is not universally accepted but is supported by the Bekenstein–Hawking entropy, the Unruh effect, and the success of horizon thermodynamics in reproducing the Einstein equations.
2. **Decoherence-produced information contributes to horizon entropy.** This is the new identification. It is consistent with the holographic principle [’t Hooft, 1993, Susskind, 1995] and Landauer’s principle, but has not been independently verified.
3. **The virial coupling $\beta = \Omega_m/2$ is exact.** The virial theorem applies to systems in equilibrium. Cosmological structure formation is a non-equilibrium process, and the effective coupling may differ from the equilibrium value. The companion observational paper tests this assumption by allowing μ_0 to float.
4. **The activation function $\mathcal{E}(a) = \exp(1 - 1/a)$ is exact.** This follows from the cumulative integral and the exponentiation of the partition function. The functional form is approximate; the MGCAMB parameterization $\mu(a) = 1 + \mu_0 \Omega_{\text{DE}}(a)$ approximates the exact form to within 1–2.5%.
5. **Perturbation-level only.** This paper derives the modification to the gravitational coupling governing linear perturbations. The full framework also predicts a background-level modification to the Friedmann equation (Eq. 21), which requires a dual-sector implementation within a Boltzmann solver and is deferred to future work.

8 Conclusion

We have presented a physical argument connecting quantum decoherence, Landauer’s principle, and horizon thermodynamics to the phenomenological μ – Σ framework for modified gravity. The argument identifies a single new physical input—the informational entropy from gravitational decoherence—and derives a specific, testable prediction: $\mu_0 = -0.135$, $\Sigma = 1$.

The dual-sector structure ($\mu \neq 1$, $\Sigma = 1$) arises from the causal structure of spacetime: degrees of freedom on timelike worldlines undergo decoherence and therefore experience modified gravity, while degrees of freedom on null worldlines do not decohere and therefore experience standard GR.

The arrow of time, the production of classical information, and the modification of gravity are aspects of the same physical process.

A companion paper [Mahaffey, 2026] demonstrates that the prediction is consistent with Planck 2018 and large-scale structure data. Upcoming surveys including Euclid and DESI will provide a definitive test of the predicted amplitude.

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Data Availability

The MCMC chains and analysis scripts supporting the companion observational paper are publicly available at:

<https://github.com/hmahaffeyges/IAM-Validation>

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